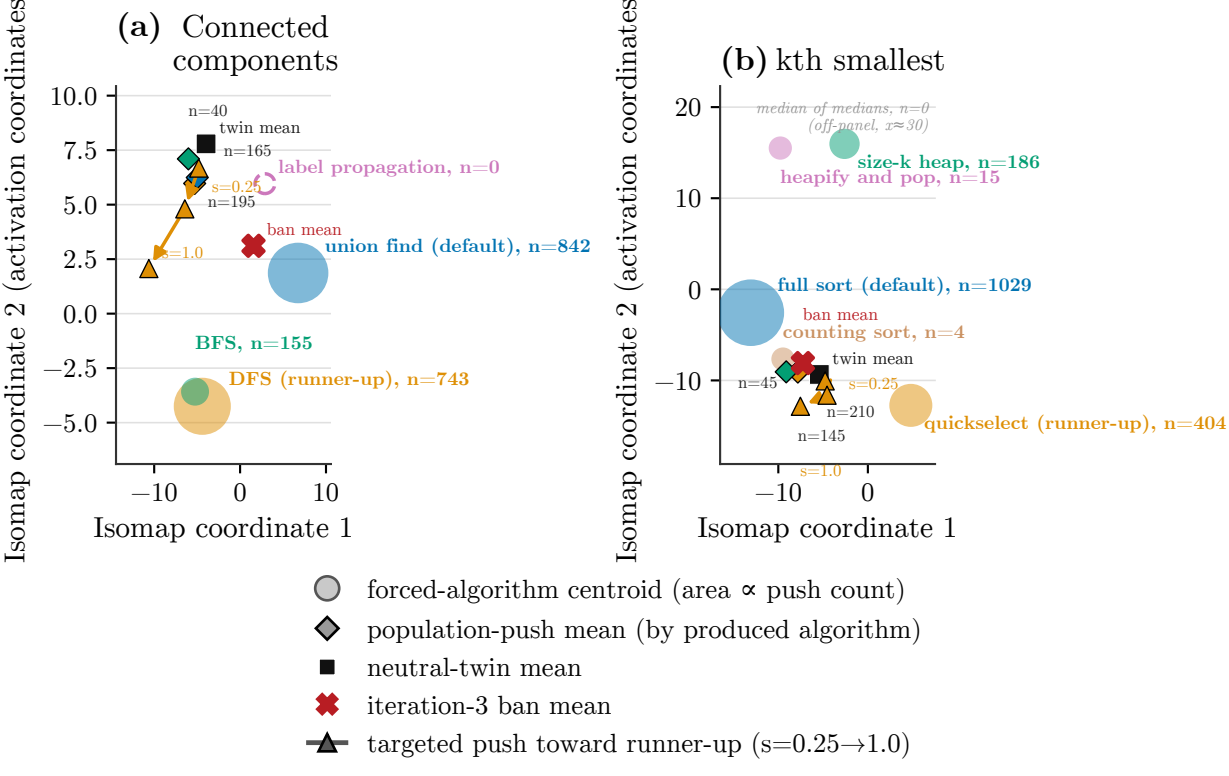
(a) Connected components
Isomap coordinate 2 (activation coordinates)
Isomap coordinate 1
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−10
0
10
n=40
twin mean
n=165
label propagation, n=0
s=0.25
n=195
ban mean
union find (default), n=842
s=1.0
BFS, n=155
DFS (runner-up), n=743
(b) kth smallest
Isomap coordinate 2 (activation coordinates)
Isomap coordinate 1
20
10
0
−10
−10
0
median of medians, n=0
(off-panel, x≈30)
size-k heap, n=186
heapify and pop, n=15
full sort (default), n=1029
ban mean
counting sort, n=4
twin mean
n=45
s=0.25
n=210
n=145
quickselect (runner-up), n=404
s=1.0
forced-algorithm centroid (area ∝ push count)
population-push mean (by produced algorithm)
neutral-twin mean
iteration-3 ban mean
targeted push toward runner-up (s=0.25→1.0)